Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

ML engineer and data scientist with **8+ years of production systems experience and a full ML model lifecycle track record**. Built **Isolation Forest anomaly detection** at GMU (cut manual QA 70%); **Random Forest + Logistic Regression on 104K records at 85%+ accuracy** (CrashClarity); **XGBoost risk segmentation** on 20+ variables. Built **AgentProbe's 5-metric eval engine with CI/CD integration and regression drift monitoring** - the same model lifecycle discipline required here. Expert in Python, SQL, PyTorch, scikit-learn, and XGBoost. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

CORE COMPETENCIES

ML Model Lifecycle | Fraud Pattern Detection | Python | SQL | PyTorch | scikit-learn | XGBoost | Isolation Forest | Feature Engineering | Model Monitoring | Data Drift Detection | AWS (S3, EC2, Glue)

EXPERIENCE

George Mason University
Aug 2024 - May 2026

Research Associate - Applied ML & Anomaly Detection

Fairfax, Virginia

- Built **Isolation Forest anomaly detection pipeline** (Python, scikit-learn) across 3 research pipelines - surfaced behavioral anomalies that manual expert review missed, **cut manual QA time by 70%**.
- Owned full ML lifecycle: built regression models (**$R^2 = 0.977$**) in Python and R across WHO + UK Road Safety datasets (190+ countries) - data collection, feature engineering, model selection, validation, and stakeholder dashboards.
- Designed **Tableau, ggplot2, and Quarto dashboards** for faculty leadership - improved research comprehension across two cohorts by **40%**.

Entrata India (Db Xento Systems)
Oct 2018 - Aug 2024

Software Engineer - Production Data Pipelines & ML Systems

Pune, India

- Applied **Python, PyTorch, and OpenAI API** to automate data transformation across **1.6M financial records** - production ML on regulated data with output validation, quality gates, and error escalation.
- Achieved **100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines - **zero security incidents**; built 5+ internal tools cutting resolution time by 40%.
- Cut deployment times by **40%** by building 20+ production microservices (Python, PostgreSQL, RabbitMQ) across 120+ Agile sprints - **100K+ MAU, 99.5% uptime** sustained over 6+ months.
- Tuned **200+ SQL queries** for a **30% production performance gain**; containerized 30%+ of microservices with Docker (50% execution time reduction).
- Eliminated **80% of manual operational workload** via Python/PL-SQL automation; conducted 300+ code reviews enforcing TDD/BDD and secure coding standards.

Entrata India (Db Xento Systems)
Jul 2017 - Sep 2018

Associate Software Engineer

Pune, India

- Improved system stability by **35%** building 20+ full-stack modules (Python, PostgreSQL) - 99.5% uptime and 25% faster report generation.

PROJECTS

CrashClarity - ML Pipeline Ownership (104K Records, 85%+ Accuracy)

2025 | Python, R, scikit-learn, Random Forest, Logistic Regression, Feature Engineering | Live: Portfolio

- Owned full lifecycle: problem framing, feature engineering across weather/road/time variables, model selection (Random Forest vs. Logistic Regression), evaluation (precision/recall/F1), and stakeholder deployment - **85%+ accuracy on 104K+ records**.

AgentProbe - Model Monitoring, Drift Detection & Eval Automation In Progress

2024-Present | Python, LangChain, LangGraph, FastAPI, ChromaDB, PyTorch, scikit-learn | github.com/venkatalakshmiparimala/AgentProbe

- Built **5-metric evaluation engine** (accuracy, hallucination, retrieval, completeness, compliance) with CI/CD integration - failing model outputs blocked automatically before production; **regression drift tracked** per run via ChromaDB.
- Designed 3-agent LangGraph runtime and **15-endpoint FastAPI backend** with multi-provider model routing (Anthropic, OpenAI, Groq, Ollama) - 100% local deployment at zero cost.

Global Health & Road Safety - Anomaly Detection System (Isolation Forest)

2024-2025 | Python, R, Isolation Forest, scikit-learn, Tableau | George Mason University

- Built **unsupervised anomaly detection system (Isolation Forest)** - flagged behavioral deviations from expected patterns across 3 active pipelines; cut manual QA by **70%**. Same core detection methodology as fraud signal detection.

EDUCATION

M.S. Data Analytics Engineering - George Mason University — 2024-2026

GPA: 3.87 · Virginia, USA · ML systems, statistical modeling, anomaly detection, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering — 2013-2017

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript, Bash

ML & Detection: PyTorch, TensorFlow, scikit-learn, XGBoost, Isolation Forest, Random Forest, Logistic Regression, Feature Engineering, Statistical Modeling, NumPy, Pandas, BERT, spaCy, Hugging Face

MLOps & Infra: AWS (S3, EC2, RDS, Glue), Docker, FastAPI, Flask, PostgreSQL, MySQL, RabbitMQ, GitHub Actions CI/CD, ChromaDB

AI & Agents: LangChain, LangGraph, OpenAI API, Anthropic Claude API, LLM Router, RAG Pipelines, Prompt Engineering **Visualization:** Tableau, Power BI, ggplot2, Plotly, Quarto

Process: TDD/BDD, Git, JIRA, Agile/Scrum, secure coding, PHI/HIPAA compliance